A SEPARATED FINITE-TYPE SCHEME OF DIMENSION AT MOST 1 OVER A FIELD IS H-QUASI-PROJECTIVE

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Allowed standard prerequisites (editorial).

- Dim-1 Noetherian separated schemes have ample invertible sheaves. delivered companion proposition-dim-1-noetherian-separated-has-ample. First input.
- Finite type with ample invertible sheaf is quasi-projective. Morphisms, Lemma morphisms-lemma-quasi-projective-finite-type-over-S. Second input.

Source reference. Original proposition 09NZ: sources/stacks-chow/tags.txt, source lines 12232–12232. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01VS: sources/stacks-chow/tags.txt, source lines 2457–2457. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01VW: sources/stacks-chow/tags.txt, source lines 2461–2461. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let X be a separated finite type scheme over a field k. If $\dim(X) \leq 1$ then X is H-quasi-projective over k.*

Proof. By Proposition 09NZ the scheme X has an ample invertible sheaf $\mathcal{L}$. By Morphisms, Lemma 01VS we see that X is isomorphic to a locally closed subscheme of $\mathbf{P}^n_k$ over $\mathrm{Spec}(k)$. This is the definition of being H-quasi-projective over k, see Morphisms, Definition 01VW. □

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